



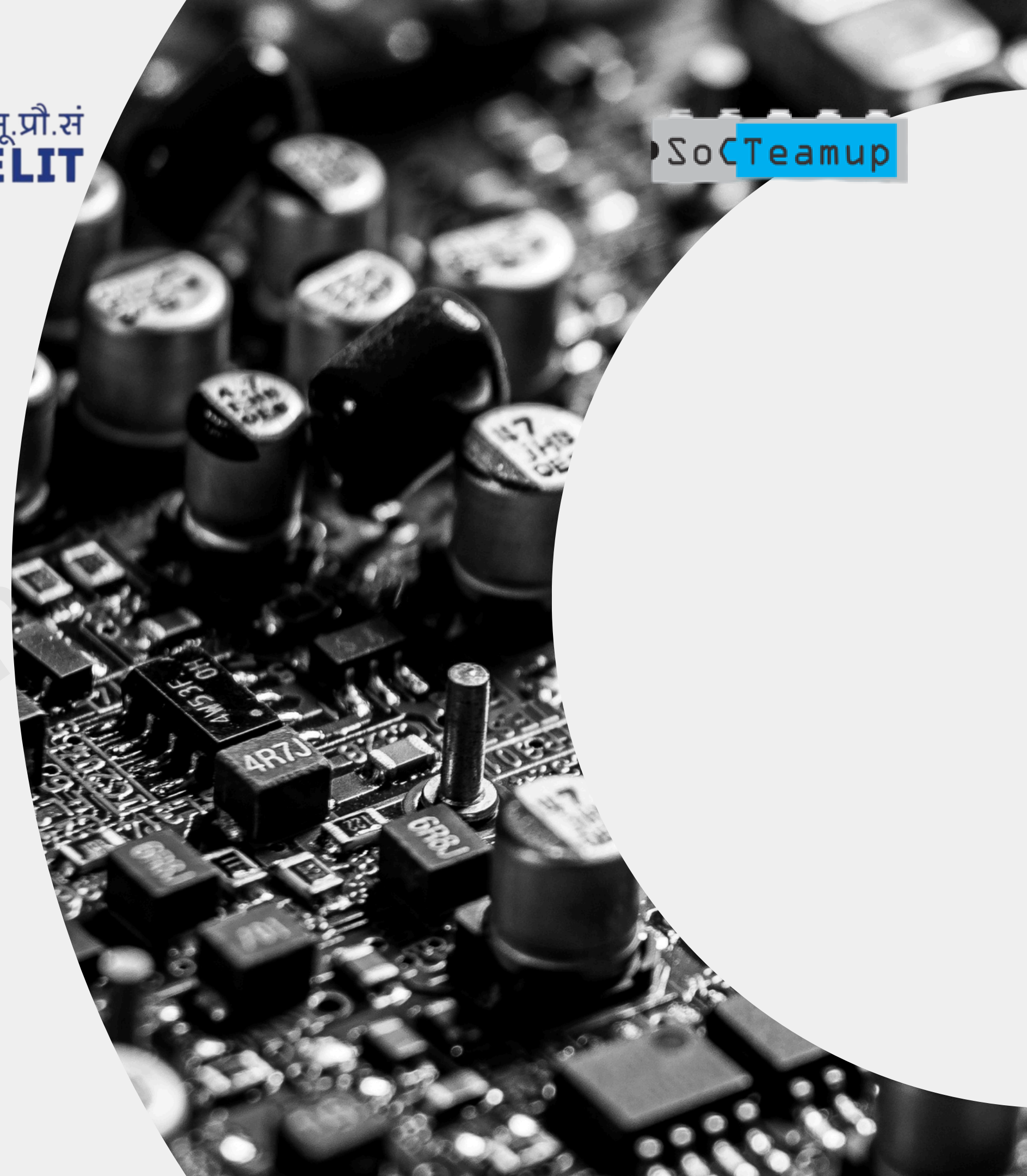
इलेक्ट्रॉनिक्स एवं  
सूचना प्रौद्योगिकी मंत्रालय  
MINISTRY OF  
ELECTRONICS AND  
INFORMATION TECHNOLOGY



SoC Teamup

# Introduction to Yosys

echiph







# Outline

**What is Yosys?**

**Merits of Yosys**

**Supported input formats**

**Basic Yosys flow**

**Key commands**

**Example Verilog  
synthesis**

# What is **Yosys**?

- **Yosys = Yosys Open Synthesis Suite**
  - Open-source RTL synthesis tool for digital hardware design
  - Converts Verilog into a gate-level netlist

# ➔ What is **Yosys**?

- **Yosys = Yosys Open Synthesis Suite**
  - Supports ASIC and FPGA flows
  - Written in C++ with modular architecture
  - Works well with open-source EDA tools (like nextpnr, Verilator)



## Merits of **Yosys**?

- Free and open-source (no licensing cost)
- Lightweight and cross-platform
- Highly scriptable (batch mode)
- Great for learning synthesis concepts

## Merits of **Yosys**?

- Used in real projects — e.g., RISC-V cores, LiteX SoCs
- Supports various FPGA families:  
Lattice iCE40, ECP5, Xilinx 7-series  
(via open-source flows)



# Supported Input **Formats**

- Synthesizable Verilog (IEEE 1364-2005)
- Some support for System Verilog (limited subset)
- RTLIL (Yosys internal format)

# **Basic Yosys Flow**

- **Read Verilog**
  - `read_verilog mydesign.v`
- **Check the hierarchy of the design from the top**
  - `hierarchy -check -top <module_name>`
- **Internal Processes and fsm, memory mapping with optimizations**
  - `proc; opt; fsm; opt; memory; opt`

## **Basic Yosys Flow**

- **Mapping Design to internal cell library**
  - techmap; opt
- **Mapping Flip Flops to standard cell library**
  - dfflibmap -liberty  
standard\_cell\_library.lib

(Eg: dfflibmap - liberty /home/<coeno>/Desktop/Technology\_PDKs/osu018/osu018\_stdcells.lib)



## **Basic Yosys Flow – Contd.**

- **Mapping logic to standard cell library**
  - abc –liberty standard\_cell\_library.lib
  - (Eg: abc –liberty /home/<coeno>/Desktop/Technology\_PDKs/osu018/osu018\_stdcells.lib)
- **Clean your design**
  - clean
- **Open the pictorial representation of the design**
  - show
- **Writing synthesized gate-level netlist**
  - write\_verilog my\_design\_synth.v

## Key Commands

- **read\_verilog** – Load Verilog file
- **hierarchy** – Check module hierarchy
- **proc** – Process always blocks
- **opt** – Optimize logic
- **techmap** – Map to technology cells
- **abc** – Technology mapping and logic optimization
- **write\_\*** – Export netlists in various formats

# ➔ Example: Synthesizing a Counter

- yosys
- read\_verilog counter.v
- hierarchy -check -top counter
- proc; opt; fsm; opt; memory; opt
- techmap; opt
- dfflibmap -liberty /home/<coe-  
no>/Desktop/Technology\_PDKs/osu  
018/osu018\_stdcells.lib

# ➔ Example: Synthesizing a Counter

- `abc -liberty /home/<coeno>/Desktop/Technology_PDKs/osu018/osu018_stdcells.lib`
- `clean`
- `show`
- `write_verilog counter_synth.v`



# Thank You

echiphu

